

# Simran Nerval — Curriculum Vitae

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🌐 <https://simrannerval.github.io/>

## Education

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### University of Toronto

*PhD in Astronomy and Astrophysics*

**Toronto, ON, Canada**

2021 – 2026 (Expected)

- Thesis: "Multi-pronged Approach to Characterize and Constrain the Early Universe" *Supervised by Renée Hložek*

### Queen's University

*MSc in Physics, Engineering Physics & Astronomy*

**Kingston, ON, Canada**

2019 – 2021

- Thesis: "Stochastic Gravitational Wave Backgrounds from Low-Scale Inflation" *Supervised by Joseph Bramante*

### University of Toronto

*Honours Bachelor of Science (Physics & Astronomy Specialist, Mathematics Minor)*

**Toronto, ON, Canada**

2015 – 2019

- Thesis: "Analysis of LiteBIRD Systematics and their Impact on Cosmology" *Supervised by Renée Hložek*

## Awards

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### Queen Elizabeth II Graduate Scholarship in Science and Technology

*CAD\$15,000, Merit-based scholarship awarded to students in science and technology.*

**University of Toronto**

2024 – 2025

### Walter C. Sumner Memorial Fellowship

*CAD\$6,650 for 2023 – 2024, CAD\$7,400 for 2024 – 2025,*

*Awarded to 40 graduate students across Canada/year*

**Walter C. Sumner Foundation**

2023 – 2025

### Dunlap Seed Funding

*CAD\$6,000 for Age of the Universe and CAD\$30,000 for Coding the Cosmos*

*Awarded to run coding and astronomy camps for low socioeconomic high school students.*

**Dunlap Institute**

2023 – 2024

### Lachlan Gilchrist Fellowship

*CAD\$4,000, Awarded to 3 graduate students in the physical sciences/year*

**University of Toronto**

2022 – 2023

### Natural Sciences and Engineering Research Council of Canada (NSERC) Scholarships

*PGS D - CAD\$21,000/year for the first 2 years, CGS D - CAD\$35,000/year for the final year*

**NSERC**

2021 – 2024

### David A. Dunlap Department of Astronomy & Astrophysics Entrance Scholarship

*CAD\$5,000*

**Uni. of Toronto**

2021

### Arts '49 Principal Wallace Fellowship

*CAD\$20,000, Awarded to 1 graduate student/year*

**Queen's University**

2020 – 2021

## Supervision, Teaching, and Mentorship Experience

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### Supervision

- Emma Xu (2024 – Present): Co-supervising Astronomy and Physics undergraduate who is utilizing numerical simulations to determine gravitational wave signatures from preheating and modifying the code to also simulate the signals during inflation.
- Ezra Msolla (2024 – 2025): Co-supervised Astronomy and Physics undergraduate who was developing an emulator to enable more efficient sampling of inflationary particle burst models.

- Zeling Zhang (2023 – 2025): Co-supervised Computer Science undergraduate who developed the back-end and a more efficient labelling system for our glitch classification project on Zooniverse enabled by active learning.

## Teaching.....

### Teaching Assistant

**Toronto, ON, Canada**

*Department of Astronomy & Astrophysics, University of Toronto*

2021 – 2025

- AST222: Ran and made materials for tutorials, held office hours, marked, created coding based problem set questions, and final exam questions for a second year galaxies and cosmology course for astronomy and physics students (January – April 2023, January – April 2024, and January – April 2025).
- AST325: Ran and made materials for astronomy and Python based tutorials, held office hours, ran labs sessions, and marked for a third year practical astronomy course for astronomy and physics students (September – December 2023 and September – December 2024).
- AST101: Ran tutorials, marked, and ran observing nights for a first year course for non-science students about the Sun, planets, comets, and the formation of the solar system (September – December 2021 and September – December 2022).

### Teaching Assistant

**Kingston, ON, Canada**

*Department of Physics, Engineering Physics & Astronomy, Queen's University*

2019 – 2020

- APSC111 (September – December 2019) and APSC112 (January – April 2020): Conducted tutorials and marked for a first year classical mechanics/electricity and magnetism course for engineering students.

## Mentorship.....

### STEM Leaders: Launch Program

**Toronto, ON, Canada**

*Mentor, Visions of Science*

2025

- Co-supervised high school student on a research project where they used ACT and SDSS data to determine the Hubble constant with Python.
- Mentored the co-supervisor on how to supervise students, develop a project plan, and develop materials that help teach the student while still allowing them to have freedom to make their own research choices.

### Graduate and Undergraduate Mentorship Programs

**Toronto, ON, Canada**

*Mentor, Department of Astronomy & Astrophysics, University of Toronto*

2021 – Present

- Mentoring upper year Physics and Astronomy undergraduates Daniella Morrone (2021 – 2022), Kaitlin Cranston (2023 – 2024), Camille Paule (2024 – Present), and Maëlle Magnan (2024 – Present).
- Mentored first year Astronomy and Astrophysics graduate student Anika Slizewski (2022 – 2023).

## Outreach, Volunteering, and Service

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### Coding the Cosmos

**Toronto, ON, Canada**

*Director, Dunlap Institute for Astronomy & Astrophysics, University of Toronto*

2023 – 2025

- In this role I was the **PI** for our seed funding grants totalling CAD\$36,000, wrote curriculum, developed materials, recruited volunteers and executives, ran trainings for volunteers, oversaw and managed all working groups totalling over 35 volunteers, and led organization logistics.
- **Developed** and **organized** multiple 3-day workshops for over 50 grade 9 – 11 students across Southern Ontario where they were introduced to coding, data analysis, and complete coding based astronomy projects as well as a day long workshop for 29 grade 11 and 12 students called *Age of the Universe*.

### Course and Quads Committee

**Toronto, ON, Canada**

*Committee Member, Department of Astronomy & Astrophysics, University of Toronto*

2022 – 2023

### Canadian Astronomical Society (CASCA)

**Virtual, Canada**

*Graduate Student Representative*

2020 – Present

- Queen's University Graduate Student Representative (2020 – 2021), Graduate Student Committee (2020 – 2024), Education and Public Outreach Committee (2021 – 2022), Equity and Inclusivity Committee (2021 – Present).

### University of Toronto AstroTours

**Toronto, ON, Canada**

*AstroTours Co-Director*

2021 – 2023

- Organized and ran monthly *astronomy public talks*, telescope tours, and astronomy demonstrations.

- Innovation, Diversity, Exploration, & Advancement in STEM Initiative** Kingston, ON, Canada  
*IDEAS Initiative Director of External Affairs* 2020 – 2023
- **Workshop Coordinator** for GIRLS Camp, organized and ran **STEM Stories**, and organized a joint **national symposium** with Let's Talk Science for high school students called **Let's Talk Astrophysics** on dark matter and galactic dynamics.
- Gender MINorities In Physics (GEMINI-P)** Kingston, ON, Canada  
*Mentorship Program Coordinator* 2020 – 2021
- Organized and ran a pilot **mentorship program** for upper year undergraduates with diverse mentors.
- Let's Talk Science** Kingston, ON, Canada  
*Coordinator (2020 – 2021), Volunteer (2019 – 2020)* 2020 – 2021
- As a coordinator I **managed volunteers** and connected them with outreach opportunities, organized **virtual outreach** for K-12 classes and **developed** new kits for topics such as physics, astronomy, and computer science.
  - As a volunteer I ran **classroom visits** at elementary schools involving **leading demonstrations**, giving quick lectures on the science behind the activities, and supporting students in conducting their own experiments.
- Queen's University Association for Queer Employees (QUAQE)** Kingston, ON, Canada  
*QUAQE Organizing Committee Member* 2020 – 2021
- Department of Physics, Engineering Physics & Astronomy** Kingston, ON, Canada  
*Volunteer, Queen's University* 2019 – 2021
- **Co-led** a team from the physics department to make videos and designing physics games for the virtual **Science Rendezvous Kingston 2021**, **operated a 14-inch telescope** for the general public, and volunteered at a variety of department outreach events.
- Departments of Physics and Astronomy & Astrophysics** Toronto, ON, Canada  
*Volunteer, University of Toronto* 2018 – 2019
- **Led demonstrations** for the public at a variety of department outreach events.

## Presentations

- Academic**.....
- Constraining the Early Universe with CMB Data and Gravitational Wave Backgrounds and Probing Millimeter Transients** Nottingham, England  
*University of Nottingham, Talk* July 2025
- Probing Millimeter Transients and Constraining the Early Universe with CMB Data and Gravitational Wave Backgrounds** Cardiff, Wales  
*University of Cardiff, Talk* July 2025
- Constraining the Primordial Power Spectrum with the ACT** Virtual  
*Cosmology from Home, Talk* June 2025
- Constraining the Primordial Power Spectrum with the ACT** Halifax, NS, Canada  
*Canadian Astronomical Society (CASCA) Annual General Meeting, Talk* June 2025
- The Development of Machine Learning Tools for Detecting Millimeter Sources in CMB Timestream Pre-processing** Toronto, ON, Canada  
*UofT TASTY Talk Series, Talk, S. K. Nerval and E. Hornecker* April 2025
- Millimeter Transient Detection During Timestream Preprocessing with the Atacama Cosmology Telescope** Toronto, ON, Canada  
*Canadian Astronomical Society (CASCA) Annual General Meeting, Poster* June 2024
- Millimeter Transient Detection During Timestream Preprocessing with the Atacama Cosmology Telescope** Toronto, ON, Canada  
*Hotwiring the Transient Universe VII, Talk, S. K. Nerval and E. Hornecker* May 2024

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|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Glitch Classification with the Atacama Cosmology Telescope and the Simons Observatory</b><br><i>CITA Cosmology Lunch, Talk</i>             | <b>Toronto, ON, Canada</b><br><i>December 2023</i>         |
| <b>Constraining Inflation Beyond the Standard Picture</b><br><i>Canadian Astronomical Society (CASCA) Annual General Meeting, Talk</i>        | <b>Penticton, BC, Canada</b><br><i>June 2023</i>           |
| <b>Multi-pronged Approach to Characterize and Constrain Inflation</b><br><i>University of Oslo Cosmoglobes Workshop, Talk</i>                 | <b>Oslo, Norway</b><br><i>January 2023</i>                 |
| <b>LiteBIRD and Future CMB Telescopes</b><br><i>University of Oslo Component Separation Course, Talk</i>                                      | <b>Oslo, Norway</b><br><i>September 2022</i>               |
| <b>Constraining Inflation Beyond the Standard Picture</b><br><i>University of Oslo Astronomy Lunch, Talk</i>                                  | <b>Oslo, Norway</b><br><i>August 2022</i>                  |
| <b>Stochastic Gravitational Wave Backgrounds from Low-Scale Inflation</b><br><i>International HPC Summer School, Poster</i>                   | <b>Virtual</b><br><i>July 2021</i>                         |
| <b>Gravitational Wave Backgrounds from Low-Scale Inflation</b><br><i>Canadian Association of Physicists (CAP) Congress, Talk</i>              | <b>Virtual</b><br><i>June 2021</i>                         |
| - <b>Awarded second place</b> in oral competition for the Division of Theoretical Physics                                                     |                                                            |
| <b>Gravitational Wave Backgrounds from Low-Scale Inflation</b><br><i>Phenomenology Symposium, Talk</i>                                        | <b>Virtual, Pittsburgh, PA, USA</b><br><i>May 2021</i>     |
| <b>Gravitational Wave Backgrounds from Low-Scale Inflation</b><br><i>Canadian Astronomical Society (CASCA) Annual General Meeting, Talk</i>   | <b>Virtual, Penticton, BC, Canada</b><br><i>May 2021</i>   |
| <b>Gravitational Wave Backgrounds from E- and T-Model Inflation</b><br><i>MI Annual Meeting, Talk</i>                                         | <b>Virtual, Kingston, ON, Canada</b><br><i>August 2020</i> |
| - <b>Awarded best</b> student presentation                                                                                                    |                                                            |
| <b>LiteBIRD's Projected Constraints of Inflationary Models</b><br><i>Great Lakes Cosmology Workshop, Talk</i>                                 | <b>Rochester, NY, USA</b><br><i>August 2019</i>            |
| <b>LiteBIRD's Projected Constraints of Inflationary Models</b><br><i>Canadian Astronomical Society (CASCA) Annual General Meeting, Poster</i> | <b>Montréal, QC, Canada</b><br><i>June 2019</i>            |

### Invited Outreach Presentations

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|--------------------------------------------------------------------------|--------------------------------------------|
| <b>University of Toronto AstroTours, Public Talk</b>                     | <b>Toronto, ON, February 2025</b>          |
| <b>Great Lakes Science Boot Camp For Librarians, Invited Speaker</b>     | <b>Toronto, ON, July 2024</b>              |
| <b>Visions of Science, Invited STEM Academy Speaker</b>                  | <b>Brampton, ON, April 2023</b>            |
| <b>Visions of Science, Invited Pre-recorded STEM Sparks Presentation</b> | <b>Toronto, ON, January 2022</b>           |
| <b>Queen's Observatory QUO Fast Radio Bursts, Invited Podcast Guest</b>  | <b>Kingston, ON, August 2021</b>           |
| <b>TELUS Spark Science Centre, Invited Camp Presentation</b>             | <b>Virtual, Calgary, AB, August 2021</b>   |
| <b>Royal Astronomical Society of Canada, Invited Public Talk</b>         | <b>Virtual, BC, February 2021</b>          |
| <b>Astronomy on Tap: Kingston, Invited Public Talk</b>                   | <b>Virtual, Kingston, ON, January 2021</b> |
| <b>Queen's University Observatory Open House, Invited Public Talk</b>    | <b>Kingston, ON, January 2020</b>          |

### Contributed Outreach Presentations

|                                                            |                                               |
|------------------------------------------------------------|-----------------------------------------------|
| <b>University of Toronto AstroTours, Public Talk</b>       | <b>Toronto, ON, Canada, November 2022</b>     |
| <b>IAU Communicating Astronomy with the Public, Poster</b> | <b>Sydney, NSW, Australia, September 2022</b> |

## Research Experience

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### Dunlap Institute for Astronomy & Astrophysics, University of Toronto

Supervised by Renée Hložek

Toronto, ON, Canada

September 2021 – Present

- Using a multi-pronged approach to **characterize and constrain inflation** using the cosmic microwave background and gravitational wave backgrounds.

### Arthur B. McDonald Institute (MI), Queen's University

Supervised by Joseph Bramante

Kingston, ON, Canada

September 2019 – August 2021

- Used detailed modeling and numerical simulations to **determine gravitational wave signatures** (stochastic gravitational wave backgrounds) of low scale inflationary sectors.

### Dunlap Institute for Astronomy & Astrophysics, University of Toronto

Supervised by Renée Hložek

Toronto, ON, Canada

September 2018 – August 2019

- **Modelled**  $1/f$  noise for LiteBIRD using Fisher forecasts in order to **constrain cosmological parameters**.

### Department of Physics, University of Toronto

Supervised by Kaley Walker

Toronto, ON, Canada

May 2018 – August 2018

- Executed **satellite validation** between the ACE satellite and ground based instruments in Eureka, Nunavut, Canada that measured  $\text{NO}_2$  between 2004 and 2017.

## Technical Skills

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Proficient in **Python** and have experience with **Fortran** and **Mathematica**. Comfortable using **L<sup>A</sup>T<sub>E</sub>X** and the **Linux** command line.

## Publications

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### Leading and Major Contribution.....

1. E. Calabrese, et al. (including **S. K. Nerval**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, accepted in JCAP (2025, 2503.14454)
2. **S. K. Nerval**, E. Hornecker, Y. Guan, et al., *The Atacama Cosmology Telescope: Machine Learning Driven Tools for Detecting Millimeter Sources in Timestream Pre-processing*, ApJ, Vol. 990, Issue 2, 139 (2025, 2503.10798)
3. A. Bhoonah, J. Bramante, **S. Nerval** and N. Song (alphabetical order), *Gravitational Waves From Dark Sectors, Oscillating Inflavons, and Mass Boosted Dark Matter*, JCAP, Vol. 2021, Issue 4, 43 (2021, 2008.12306)

### Collaboration.....

1. T. Louis, et al. (including **S. K. Nerval**), *The Atacama Cosmology Telescope: DR6 Power Spectra, Likelihoods and  $\Lambda$ CDM Parameters*, submitted to JCAP (2025, 2503.14452)
2. S. Naess, et al. (including **S. K. Nerval**), *The Atacama Cosmology Telescope: DR6 Maps*, submitted to JCAP (2025, 2503.14451)
3. M. Abitbol, et al. (including **S. K. Nerval**), *The Simons Observatory: Science Goals and Forecasts for the Enhanced Large Aperture Telescope*, JCAP, Vol. 2025, Issue 8, 34 (2025, 2503.00636)
4. D. J. Watts, et al. (including **S. K. Nerval**), *Cosmoglobe DR1. III. First full-sky model of polarized synchrotron emission from all WMAP and Planck LFI data*, A&A, Vol. 686, A297 (2023, 2310.13740)
5. J. R. Eskilt, et al. (including **S. K. Nerval**), *Cosmoglobe: Towards end-to-end CMB cosmological parameter estimation without likelihood approximations*, A&A, Vol. 678, A169 (2023, 2306.15511)
6. T. Hasebe, et al. (including **S. Nerval**), *Sensitivity Modeling for LiteBIRD*, J. Low Temp. Phys., Volume 211, Issue 5-6, 384-397 (2023)
7. J. R. Eskilt, et al. (including **S. K. Nerval**), *Cosmoglobe DR1 results. II. Constraints on isotropic cosmic birefringence from reprocessed WMAP and Planck LFI data*, A&A, Vol. 679, A144 (2023, 2305.02268)
8. LiteBIRD Collaboration (including **S. Nerval**), *Probing cosmic inflation with the LiteBIRD cosmic microwave*

*background polarization survey*, PTEP, Vol. 2023, Issue 4, 042F01 (2023, 2202.02773)

9. D. J. Watts, et al. (including **S. K. Nerval**), *Cosmoglobe DR1 results. I. Improved Wilkinson Microwave Anisotropy Probe maps through Bayesian end-to-end analysis*, A&A, Vol. 679, A143 (2023, 2303.08095)
10. U. Fuskeland, et al. (including **S. K. Nerval**), *Tensor-to-scalar ratio forecasts for extended LiteBIRD frequency configurations*, A&A, Vol. 676, A42 (2023, 2302.05228)
11. J. Hubmayr, et al. (including **S. Nerval**), *Optical Characterization of OMT-Coupled TES Bolometers for LiteBIRD*, J. Low Temp. Phys., Vol. 209, Issue 3-4, 396-408 (2022)
12. P. Vielva, et al. (including **S. Nerval**), *Polarization angle requirements for CMB B-mode experiments. Application to the LiteBIRD satellite*, JCAP, Vol. 2022, Issue 04, 029 (2022)
13. M. Hazumi, et al. (including **S. Nerval**), *LiteBIRD: JAXA's new strategic L-class mission for all-sky surveys of cosmic microwave background polarization*, Proceedings of the SPIE, Vol. 11443, 114432F (2020)
14. H. Sugai, et al. (including **S. Nerval**), *Updated Design of the CMB Polarization Experiment Satellite LiteBIRD*, Journal of Low Temperature Physics, Vol. 199, Issue 3-4, 1107–1117 (2020)
15. A. Lee, et al. (including **S. Nerval**), *LiteBIRD: an all-sky cosmic microwave background probe of inflation*, Astro2020: Decadal Survey on Astronomy and Astrophysics, APC white papers, no. 286; Bulletin of the American Astronomical Society, Vol. 51, Issue 7, 286 (2019)